

ADITYA ANAND

Department of Computer Science and Engineering,
Indian Institute of Technology Bombay, Mumbai, Maharashtra - 400076, India

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RESEARCH INTERESTS

My research interests are in compilers, programming languages, and program analysis. I am interested in designing precise and efficient static and dynamic program analysis techniques for just-in-time (JIT) compilers. My research focuses on combining static analysis with runtime information to enable precise and efficient optimizations in managed runtimes. In particular, my doctoral research developed a robust static+dynamic framework for optimizing object allocation in Java on a production Java Virtual Machine (JVM). I am currently focused on productizing the outcomes of my PhD research with IBM and exploring their application to production compiler and runtime systems.

WORK EXPERIENCE

Indian Institute of Technology Bombay Mumbai, India
Research Associate, Computer Science and Engineering Sept 2026 – Ongoing

EDUCATION

Indian Institute of Technology Bombay Mumbai, India
Doctor of Philosophy (Ph.D.), Computer Science and Engineering Jul 2023 – Aug 2026

- **Thesis Title:** *Precision without Regret: Sound and Efficient Staged Analysis for Managed Runtimes*
- **Advisor:** Prof. Manas Thakur
- CGPA: 9.40 / 10.00

Indian Institute of Technology Mandi Mandi, India
Doctor of Philosophy (Ph.D.), Computer Science and Engineering Sep 2020 – Jun 2023
(Transfer to IIT Bombay)

Visvesvaraya Technological University Belagavi, India
Bachelor of Engineering (B.E.), Computer Science and Engineering Aug 2015 – Jun 2019

- CGPA: 9.52 / 10.00
- Department Rank: 4th Rank out of 145 students.

PUBLICATIONS

- [1] Gauravsingh Sisodia, Poorna Teja, **Aditya Anand**, and Manas Thakur. **Beyond Java 6: Reviving Reflection-Aware Static Analysis for Modern JVMs**. In *Proceedings of the 18th ACM SIGPLAN International Workshop on Virtual Machines and Intermediate Languages, VMIL'26*, p. 8, ACM, 2026. DOI: To Appear
- [2] **Aditya Anand**, Vijay Sundaresan, Daryl Maier, and Manas Thakur. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. In *Proceedings of the ACM on Programming Languages (OOPSLA)*, Singapore, October 2025. DOI: 10.1145/3763149
- [3] **Aditya Anand** and Manas Thakur. **Partial Program Analysis for Staged Compilation Systems**. *Formal Methods in System Design (FMSD)*, Springer, June 2024. DOI: 10.1007/s10703-024-00458-x

- [4] **Aditya Anand**, Solai Adithya, Swapnil Rustagi, Priyam Seth, Vijay Sundaresan, Daryl Maier, V. Krishna Nandivada, and Manas Thakur. **Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes**. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, Copenhagen, Denmark, June 2024. DOI: 10.1145/3656389
- [5] **Aditya Anand** and Manas Thakur. **Principles of Staged Static+Dynamic Partial Analysis**. In *Proceedings of the 29th Static Analysis Symposium (SAS)*, Auckland, New Zealand, December 2022. DOI: 10.1145/3563768.3563957

Invited to appear in Formal Methods in System Design (FMSD)

POSTERS

- [P1] **Aditya Anand** et al. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. Academic Research and Careers for Students (ARCS), ACM India, IIT Hyderabad, February 2026.
- [P2] **Aditya Anand** et al. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications (OOPSLA/SPLASH Poster Track), Singapore, October 2025.
- [P3] **Aditya Anand** et al. **Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes**. Academic Research and Careers for Students (ARCS), ACM India, Coimbatore, India, February 2025.
- [P4] **Aditya Anand**. **A Study of the Impact of Callbacks in Staged Static+Dynamic Partial Analysis**. In *Companion Proceedings of ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications (SPLASH Companion, Student Research Competition)*, Auckland, New Zealand, December 2022.

HONOURS AND AWARDS

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|---|-------------|
| • Excellence in Teaching Assistantship Award, IIT Bombay | Fall 2025 |
| • George B. Fernandes Award for Excellence in Ph.D. Research Progress, IIT Bombay | 2025 |
| • ACM India / IARCS International Travel Grant | 2025 |
| • Excellence in Teaching Assistantship Award, IIT Bombay | Spring 2024 |
| • ACM India / IARCS International Travel Grant | 2024 |
| • ACM SIGPLAN Fellowship, (PLMW @ PLDI) | 2024 |
| • Best Poster Presentation Winner, RISC 24 IIT Bombay | 2024 |
| • Excellence in Teaching Assistantship Award, IIT Bombay | Fall 2023 |
| • GATE Fellowship, Ministry of Education, Government of India | 2020 |
| • 4 th Rank Holder in B.E. (Computer Science), VTU Karnataka | 2019 |
| • 1 st Place in Inter-College Technical Quiz Competition | 2018 |
| • Rajya Puraskar in Bharat Scouts and Guides | 2013 |

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member:

- Programming Language Design and Implementation (PLDI 2026).
- Object-Oriented Programming, Systems, Languages & Applications (OOPSLA 2026)
- European Conference on Object-Oriented Programming (ECOOP 2025)
- Programming Language Design and Implementation (PLDI 2024).
- European Conference on Object-Oriented Programming (ECOOP 2024)

TEACHING EXPERIENCE

- **Teaching Assistant**, CS6004: Code Optimization for Object-Oriented Languages Spring 2026
- **Teaching Assistant**, CS339+CS355: Abstractions and Paradigms for Programming Fall 2025
- **Teaching Assistant**, CS614: Advanced Compilers Spring 2025
- **Teaching Assistant**, CS339+CS355: Abstractions and Paradigms for Programming Fall 2024
- **Teaching Assistant**, CS6004: Code Optimization for Object-Oriented Languages Spring 2024
- **Teaching Assistant**, CS614: Advanced Compilers Fall 2023
- **Teaching Assistant**, CS515: Advanced Computer Science Practicum Fall 2022
- **Teaching Assistant**, CS502: Compiler Design Fall 2022
- **Teaching Assistant**, CS611: Program Analysis Spring 2022
- **Teaching Assistant**, CS302: Paradigms of Programming Spring 2021

INVITED TALKS AND RESEARCH PRESENTATIONS

1. **Invited Talk:** *Precision Without Regret: Sound and Efficient Memory Optimization in Managed Runtimes*, IIT Gandhinagar, India, March 2026.
2. **Invited Talk:** *Precision Without Regret: Efficient Optimization Strategies for JIT Compilation*, IIT Hyderabad, India, February 2026.
3. **Research Presentation:** *CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers*, Research Highlights in Programming Languages (RHPL), India, December 2025.
4. **Conference Paper Presentation:** *Combining Static Analysis and Speculation in JIT Compilers*, OOP-SLA 2025, Singapore, October 2025.
5. **Research Presentation:** *Combining Static Analysis and Speculations in JIT Compilers*, Innovations in Compiler Technology (IICT), October 2025.
6. **Research Presentation:** *Program Analysis for Managed Runtimes in Presence of Dynamic Features*, Innovations in Compiler Technology (IICT), September 2024.
7. **Research Presentation:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, Software Engineering Research in India (SERI), July 2024
8. **Invited Talk:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, ComputerSysTalks@India, June 2024.
9. **Conference Paper Presentation:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, PLDI 2024, Copenhagen, Denmark, June 2024.
10. **Research Presentation:** *Staged Static+Dynamic Partial Analysis for Java-like languages*, Software Engineering Research in India (SERI), June 2023.
11. **Conference Paper Presentation:** *Principles of Staged Static+Dynamic Partial Analysis*, Static Analysis Symposium (SAS 2022), Auckland, New Zealand, December 2022.

TECHNICAL SKILLS

- **Core Research Frameworks:** Soot (Java Program Analysis), Eclipse OpenJ9 VM, JavaCC, JTB.
- **Programming Languages:** Java, C, C++, Haskell, Scheme.
- **Systems & Systems Scripting:** Bash, Awk, Git, Docker.

ACADEMIC REFERENCES

Prof. Manas Thakur

Assistant Professor, CSE Department
Indian Institute of Technology Bombay
Email: manas@cse.iitb.ac.in
Ph.D. Research Advisor

Prof. Uday Khedker

Professor, CSE Department
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PhD Research Committee Member

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Research Project Collaborator